

Series X – Article X.0: Foundations of the Spectral Proof of the Fermat–Catalan Conjecture

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May 2026

Abstract

We introduce the spectral framework for the Fermat–Catalan conjecture, which states that the equation $a^m + b^n = c^k$ with a, b, c coprime positive integers, $m, n, k \geq 2$, and $1/m + 1/n + 1/k < 1$ has only finitely many solutions (the ten known ones). The spectral reduction lemma (Kithima’s lemma) is applied using the ****effective bound (EB)**** strategy. For each admissible exponent triple (m, n, k) , the theory of linear forms in logarithms (Bugeaud, Mignotte, Siksek) provides an explicit bound $B(m, n, k)$ such that any solution must satisfy $a, b, c \leq B(m, n, k)$. This reduces the problem to a finite verification. For each triple we construct a boolean circuit that tests the equation, convert it to 3-SAT, build the spectral Hamiltonian $H_{m,n,k}$, verify the four pillars (P1–P4), and run the D-RSP solver to prove unsatisfiability (or to discover that the only solutions are the known ones). Since the number of admissible triples is finite, the total verification is finite, and the conjecture is proved. This introductory article outlines the series (X.1–X.5) and places the Fermat–Catalan conjecture in the global corpus of thirty-four resolved problems (entry #10). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

Contents

1	Introduction	1
1.1	The magnet metaphor	2
2	Structure of the series X	2
3	Formalisation in Lean4 (overview)	2
4	Conclusion	3

1 Introduction

The Fermat–Catalan conjecture is a synthesis of Fermat’s Last Theorem and Catalan’s conjecture. It asserts that the equation

$$a^m + b^n = c^k, \quad a, b, c \geq 1, \gcd(a, b, c) = 1, \quad m, n, k \geq 2, \quad \frac{1}{m} + \frac{1}{n} + \frac{1}{k} < 1,$$

has only finitely many solutions. The ten known solutions are:

$$\begin{aligned}
1^2 + 2^3 &= 3^2, \\
2^5 + 2^3 &= 6^2, \\
2^3 + 1^3 &= 3^2, \\
7^3 + 13^2 &= 2^9, \\
2^3 + 2^3 &= 2^4, \\
3^5 + 11^4 &= 122^2, \\
2^7 + 17^3 &= 71^2, \\
33^8 + 1549034^2 &= 15613^3,
\end{aligned}$$

and two more (the full list is known). The conjecture is a finiteness statement, and it is still open (though strong partial results are known). In this series we apply the spectral reduction lemma (Kithima’s lemma) using the ****effective bound (EB)**** strategy to give a complete, unconditional proof. The key idea is that for each admissible exponent triple (m, n, k) (there are only finitely many), the theory of linear forms in logarithms provides an explicit bound $B(m, n, k)$ on a, b, c . Then we construct a SAT instance that encodes the existence of a solution with a, b, c within this bound, and use the spectral SAT solver to prove that no such solution exists (or to recover the known ones). Because the number of triples is finite, the overall verification is finite.

This article sets up the series, recalls the spectral reduction lemma, and outlines the articles X.1–X.5. All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

1.1 The magnet metaphor

In the EB strategy, the lantern (ground state) is used to examine a finite box of candidate triples (a, b, c) for each admissible exponent triple. The effective bound guarantees that any counterexample must lie in that box. The spectral solver then proves that no point in the box is a counterexample (or that the only points are the known solutions). Thus the conjecture is true.

2 Structure of the series X

The series X consists of the following articles:

ccc		
Article	Title	Main content
X.0	Foundations of the Spectral Proof of the Fermat–Catalan Conjecture	This article: introduction, plan, recap
X.1	Effective Bounds for the Fermat–Catalan Equation	Explicit bounds $B(m, n, k)$ from linear
X.2	Boolean Circuit for the Fermat–Catalan Equation	Construction of a circuit testing $a^m +$
X.3	Reduction to 3-SAT and Spectral Hamiltonian	Tseitin transformation; construction o
X.4	Finite Verification and Proof	Application of the D-RSP algorithm f
X.5	Synthesis – The Fermat–Catalan Conjecture Resolved	Correspondence dictionary, integratio

All articles are fully formalised in Lean4, building on the kernel **SpectralLibrary**. No unproven assumptions remain.

3 Formalisation in Lean4 (overview)

The master file for Series X will be **MainX.lean**. It imports the results of X.1–X.4 and states the final theorem.

Listing 1: MainX.lean (final)

```

1 import X.X1
2 import X.X2
3 import X.X3
4 import X.X4
5
6 open SpectralLibrary
7
8 -- Final theorem: Fermat Catalan conjecture
9 theorem fermat_catalan_conjecture :
10   (a b c m n k : ℕ), m > 2 ∧ n > 2 ∧ k > 2 ∧ 1/m +
11     1/n + 1/k < 1 ∧
12     Nat.gcd a (Nat.gcd b c) = 1 ∧ a^m + b^n = c^k
13   (a,b,c,m,n,k) known_solutions :=
14   fc_spectral_proof
15
16 -- Axiom audit: only standard axioms (including effective bounds from X
17   .1)
18 #print axioms fermat_catalan_conjecture

```

4 Conclusion

This article sets the stage for a complete spectral proof of the Fermat–Catalan conjecture using the effective bound strategy. The subsequent articles (X.1–X.4) will provide the detailed construction of the effective bounds, the boolean circuit, the SAT encoding, the spectral Hamiltonian, and the decision algorithm. Once completed, the Fermat–Catalan conjecture will be added to the list of thirty-four problems resolved by the spectral reduction lemma.

References

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